ITA0608 – Machine Learning Lab Practicals

EXP 10:

Write a Python Program to Implement Expectation & Maximization Algorithm.

Aim:

To implement the Expectation Maximization (EM) algorithm in Python and demonstrate clustering using an appropriate dataset.

Algorithm:

- Load the dataset and split it into feature variables.
- Create a Gaussian Mixture Model (GMM) with the required number of clusters.
- Train the model using the Expectation Maximization (EM) algorithm.
- Predict the cluster labels for all data samples.
- Display the cluster assignments and evaluate the clustering results.

Program:

```
from sklearn.datasets import load_iris

from sklearn.mixture import GaussianMixture

from sklearn.metrics import accuracy_score

import pandas as pd

iris = load_iris()

X = iris.data

y = iris.target

gmm = GaussianMixture(

    n_components=3,

    random_state=42

)

gmm.fit(X)

clusters = gmm.predict(X)

result = pd.DataFrame({

    "Actual Class": y,

    "Predicted Cluster": clusters

})

print("First 10 Samples:\n")
```

```
print(result.head(10))
print("\nCluster Centers:\n")
print(gmm.means_)
accuracy = accuracy_score(y, clusters)
print("\nApproximate Accuracy:", round(accuracy * 100, 2), "%")
```

Output:

```
First 10 Samples:

   Actual Class  Predicted Cluster
0             0                  1
1             0                  1
2             0                  1
3             0                  1
4             0                  1
5             0                  1
6             0                  1
7             0                  1
8             0                  1
9             0                  1

Cluster Centers:

[[6.54639415 2.94946365 5.48364578 1.98726565]
 [5.006      3.428      1.462      0.246     ]
 [5.9170732  2.77804839 4.20540364 1.29848217]]

Approximate Accuracy: 0.0 %
```

Result:

The Expectation Maximization (EM) algorithm was successfully implemented in Python using the Gaussian Mixture Model (GMM) on the Iris dataset.